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Prevalence of anti-hepatitis C cirus antibody among pregnant women and blood donors at Bowen University Teaching Hospital, Ogbomoso, Oyo State, Nigeria

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Abstract

Hepatitis C virus is one of the emerging infectious diseases that can be transmitted through blood-to-blood contact. This study was carried out to determine the prevalence of anti-HCV antibodies among potential blood donors and pregnant women attending Bowen University Teaching Hospital (BUTH), Ogbomoso, Oyo State. Hospital based study was conducted from December 2014 to September 2015. The study group (N = 279) included potential blood donors and pregnant women. Data on socio-demographic characteristics and potential risk factors were collected using a structured questionnaire. The presence of anti-HCV antibodies in serum samples of the studied subjects were detected using third generation Enzyme Linked Immunosorbent Assay (ELISA) (WKEA Med Supplies Corp, China). Chi-square test was utilized to assess the association between the socio-demographic variables and HCV status. Logistic regression was done to determine the strength of association between risk factors and HCV status. Statistical significance was set at $P < 0.05$. Overall seroprevalence of hepatitis C virus infection was found to be 1.79% consisting 0.36% of pregnant women and 1.43% of blood

donors. None of the socio-demographic characteristics and potential risk factors among the study groups were significantly associated with hepatitis C virus infection. This study found a seroprevalence of anti-HCV antibody to be 1.79%, thus, screening of pregnant women and blood donors for HCV infections with the use of ELISA is recommended because of its important role in detecting the presence of anti-HCV antibody with utmost specificity and sensitivity. HCV, Pregnant women, Blood donors, Seroprevalence, Socio-demographic factor, BOWEN, Ogbomoso.

Introduction

Hepatitis C Virus (HCV) is a single-stranded, positive-sense RNA virus, belonging to the *Hepacivirus* genus in the family *Flaviviridae*. Its inclusion in the family *Flaviviridae* is based upon a number of similarities in its genome organization, structure and replication to a large group of vector-borne viruses causing diseases such as dengue fever and yellow fever [1, 2]. It is mainly transmitted by parenteral route; however although with lower efficacy, it may also be transmitted by sexual intercourse and by the mother to child route [3].

HCV infection is often asymptomatic, but chronic infection can lead to scarring of the liver and ultimately to cirrhosis, which is generally apparent after many years. In some cases cirrhosis leads to liver failure, liver cancer, or life-threatening oesophageal and gastric varices [4, 5]. The geographic distribution and genetic divergence among HCV genotypes may provide important indications not only about the origin but the length of time that the virus has been present in a region [6].

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Approximately 500 000 people die each year from hepatitis C-related liver diseases [7]. The actual prevalence of HCV is difficult to assess because serological tests do not discriminate among acute, chronic, or resolved infections and the analyzed serum in most countries are not representative of the general population, such as blood donors, pregnant women, drug abusers, or individuals with high-risk sexual practices. However, countries in Africa and Asia historically have the highest reported prevalence [8, 7]. According to Halim and Ajayi [9], it is estimated that HCV prevalence in Nigeria ranges between 5.8% and 12.3%.

HCV is plasma-borne and has the same routes of transmission in comparison with hepatitis B virus - sexual contact, exposure to contaminated blood and blood products, organ transplants, or vertical transmission, that is, from mother to her foetus during the prenatal period, although it occurs in less than 10% pregnancies [10, 4]. HCV may spread among blood donors and pregnant women due to vertical transmission and sexual abuse. Early detection of this virus in both pregnant women and blood donors is of immense value in disease management. The use of ELISA in the detection of hepatitis C viral infection in a population has provided reliable result as it has utmost specificity and sensitivity when compared to test strips and other kits.

HCV infection is of global concern as the prevalence is increasing in the developing countries including Ogbomoso. The seroprevalence of HCV among blood donors and pregnant women with the use of ELISA is less documented in Ogbomoso and Nigeria, justifying this research. There is an urgent need for this research to provide necessary information for proper management of the infection in the study area.

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Materials and methods

Study Area

The study was carried out at the Bowen University Teaching Hospital situated at Ogbomoso, Oyo State in the tropical belt of Southern Western part of Nigeria (Africa).

Study Population

Subjects for the seroprevalence study were recruited from any potential donor at the phlebotomy section of Bowen Teaching Hospital blood bank and pregnant mothers attending antenatal clinic. Questionnaires were administered to all participating subjects as well as verbal and informed consent form prior blood sample collection.

Ethical Approval

Ethical approval for the study was obtained from the Ethics Review Board of Bowen University Teaching Hospital, Ogbomoso. Informed consent was obtained from all subjects recruited into the study.

Exclusion/Inclusion Criteria for Subjects

Apparently potential blood donor, between 18 to 56 years of age, at the donor section of BUTH blood bank and pregnant mothers attending antenatal clinic were recruited for this study. There was no gender exclusion among the potential blood donor subjects. Those individual who

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refused to give consent as well as provide necessary information on the questionnaire were excluded from the study.

Sample Size Determination

The sample size was determined using Fischer's formula for Base Sample - Size Calculation and it has been calculated to be 276. The sample size of the proposed study was calculated using 0.24 prevalence and a minimum sample size of 279 was obtained and included in the present study

$$n = \frac{t^2 \times P(1-P)}{m^2}$$

Where n = required sample size, t = confidence level at 95% (1.96), p = estimated prevalence in the project area (0.24, using Ojo *et al*., 2013), m = margin of error at 5% (0.05).

Collection of Blood Sample

Five ml of blood was collected from potential blood donors as well as from pregnant women attending BUTH, dispensed into a plain container and allowed to clot for separation.

Processing of Blood Sample

The clotted specimen was centrifuged at 1000 RCF (relative centrifugal force) for ten minutes, the serum was picked into 2 cc labelled cryovials using sterile pipette and stored at -20ºC until ready for assay. The serum was used for detection of anti-HCV antibodies.

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Anti-HCV antibody detection

Commercially available third generation ELISA kits (WKEA MED SURPLUS CORP, China) were employed for the detection of HCV antibodies according to the manufacturer's instructions. The optical density was read using the Emax endpoint ELISA microplate reader (Molecular Devices, California, USA) and the result was interpreted according to the manufacturer's instruction.

Statistical Analysis

The prevalence of HCV infection was determined from the proportion of the positive individuals in the total population studied and expressed as percentages. The data obtained was analyzed using a Statistical Package for Social Sciences (SPSS version 20). They were organized and summarized in terms of frequencies and the results of the study were presented in tables. Chi-square (χ^2) test was utilized to assess the association between the socio-demographic variables and HCV status. Statistical significance was set at $P < 0.05$. Logistic regression was done to determine the strength of association between the potential risk factors and anti-HCV antibodies.

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Results

Social demographic characteristics of the study participants

A total of two hundred and seventy nine (279) individuals, comprising of 138 (49.8%) pregnant women and 141 (50.2%) blood donors consented for the study.

Among the 141 blood donors included in this study, 123 (87.23%) were males while 18 (12.77%) were females. The mean age of the study subjects was 31.13 years (SD ± 9.73) with majority falling in the age range of 16-25 years (43.26%). The lowest number was in the range of 56-65 years (0.71%). Majority of the subjects were single (55.32%) followed by monogamy (39.01%) and polygamy (5.67%). Of participants, 18 (12.77%) did not attend formal education while 68 (48.23%) had tertiary education. Among the 141 blood donors, 55 (39.01%) were students while 4 (2.84%) were others (Table 1).

Of 138 pregnant women included in this study, the mean age was 31.40 years (SD ± 10.82) with majority of the women falling in the age category of 26-35 years. This age category constituted 68.12% of the total pregnant women and the lowest number was in the age range 16-25 years (15.22%). Majority of the subjects were monogamy (95.65%) followed by polygamy (4.35%). Among the 138 participants, 48 (34.78%) were trading followed by 46 (33.33%) civil servants and 1 (0.72%) was in the other category. Majority of the participants (99.28%) were not aware or vaccinated of/for HCV. 66 (55.07%) of the participants had tertiary education while 2 (1.45%) had no education. 57.97% had parity of between 1 and 3 (Table 1).

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Association of anti-HCV antibodies and socio-demographic characteristics

As shown in Table 2, out of 141 blood donors, 4(2.84%) individuals were positive for HCV antibody while among 138 pregnant women, 1(0.72%) was positive for HCV antibody.

Amongst the blood donor, the highest age-specific prevalence (4.92%) was found in 16-25 years age group followed by (4.35%) in 36-45 years age group. There was no statistically significant association between reaction of anti-HCV antibody and the age categories ($P = 0.590$). From 141 blood donors, 87.23% were males out of which 3.25% were positive for HCV antibody. There was no statistically significant association between anti-HCV antibody and sex distribution of the study subjects ($P = 0.557$). Majority of the blood donors were single (55.32%), of which 3.85% were positive for HCV antibody. 1.82% was positive in the monogamy group. There was no statistically significant association between anti-HCV antibody and the marital status of the study subject ($P = 0.699$). Occupation wise, 3.64% of students and 8.70% of those trading were positive for HCV antibody while 2.17% of civil servant was positive for HCV antibody. In this study, no statistically significant association was observed between anti-HCV positivity and occupation of the blood donors ($P = 0.305$). The highest prevalence of HCV antibody in the educational categories were 14.71% for tertiary, 2.56% for secondary and 11.11% for none among the participants (Table 3). However, there is no statistically significant association between anti-HCV antibody distribution and educational status ($P = 0.171$) (Table 3).

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Amongst the pregnant women, as shown in Table 4, the age-specific prevalence was identified in the age group 36-45 (4.35%), however, there was no statistical significance between the age groups and HCV positivity (P = 0.150). Out of 95.65% of the participant in the monogamy group, 0.75% was positive to HCV antibody. There was no statistical significant association between marital status and anti-HCV antibody (P = 0.831). Occupation wise, 2.17% of the civil servants was positive to HCV antibody. There was no statistical significant association between occupation distribution of the study subject and anti-HCV reaction (P = 0.847). The prevalence of HCV antibody in the educational category was 1.32% out of 76 participants in the tertiary group. Parity-specific prevalence was identified in the multigravida group (2.04%), however, there was no statistical significant association between parity status and anti-HCV reaction (P = 0.412).

Discussion

HCV infections are significant health problems around the globe. It is associated with a broad range of clinical presentations ranging from acute hepatitis to chronic infection that may be clinically asymptomatic or may progress to chronic hepatitis and liver cirrhosis [11].

The overall prevalence of HCV in this study was 1.79% (5/279). This result was nearly similar to that obtained by Babatunde *et al*., [12] and Nwanko *et al*., [13] in Nigeria. In other Sub-Saharan African countries, 5.8% has been reported from Ethiopia [14], 2.4% from Zambia [15], 7.6% and 8.3% from Egypt [16, 17], 8.4% from Ghana [18], 5.0% from Central African Republic [19], 8.0% from Tanzania [20], suggesting that the value obtained from this study is much lower,

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whereas, it was higher than 0.26% reported from Ethiopia [11], 0.6% from Northwest Ethiopia [21], 0.9% from Namibia [22] and 1.1% from Sudan [23].

In Nigeria, the prevalence observed in this study is lower than a 2% from Sokoto [24], 2.4% from Kano [25], 2.9% from Niger Delta [26], 4.9% from Jos [27], 5.8% from Southern Nigeria [28], 6% prevalence from Osogbo [29], 6% from Jos [30], 12.3% from Benin [9], however, it is slightly higher than a 0.5% prevalence from Port Harcourt [24] and 1.5% from Lagos [31]. These differences in the seroprevalence may be due to sample size, increased awareness, less exposure to risk factors and difference in the sensitivity and specificity of test kits.

Low HCV prevalence recorded among pregnant women (0.36%) as compared with blood donors (1.43%) is consistent with the findings of Ejele *et al*., [26] and Achinge *et al*., [32].

The seroprevalence of 1.43% was obtained for anti-HCV among the blood donors (Table 2). In Sub-Saharan Africa countries, this result is lower than the reports of 8.0% by Mantee *et al*., [20] in Tanzania, 8.4% by Ampofo *et al*., [18] from Ghana, 5.8% from Ethiopia [14], 5.0% from Central African Republic [19] and 7.6% from Egypt [16] but higher than the report of Buseri *et al*., [29] who reported 1.1% in Osogbo, Nigeria.

Among the blood donors, socio-demographic variables like age, sex, marital and educational status, occupations of the participants were not significantly associated with the risk of HCV infection ($P > 0.05$) (Table 3). This result is consistent with the reports by Obienu *et al*., [33], Mboto *et al*., [34], Achinge *et al*., [32] and Adejare *et al*., [31]. As shown in Table 1, the majority of the blood donors were male (87.23%). This corroborates with the study of Adejare *et*

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al., [31], Egah *et al*., [30], Muktar *et al*., [35] and Malhotra *et al*., [36]. The distribution of HCV antibody according to sex in this study shows that male had higher prevalence (3.25%) when compared with females (0.0%). This observation is in line with the findings of Inyama *et al*., [37] and Afrose *et al*., [44], however, it disagrees with the reports of Ejele *et al*., [26] and Sule *et al*., [38] where females had higher prevalence than males. Statistical analysis showed no significant difference ($P > 0.05$) between the prevalence rate and gender distribution. The highest prevalence was in the age group 16-25 (4.92%) years followed by 36-45 (4.35%) years, however, there was no statistical significance between age and positivity of HCV antibodies ($p = 0.590$). This finding is in agreement with previous results of Erhabor *et al*., [24] who reported a high prevalence of HCV antibodies among the age range 18-43 years which are among the age groups in this study. Considering this age group, high prevalence of HCV antibodies might be as a result of sexual activity, intravenous drug use and abuse, tattooing and other unhygienic activities. The prevalence of HCV antibodies was higher among the single donors 3(3.85%) followed by the married (monogamy) donors 1(1.82%), however, there was no statistical significance ($P = 0.699$). This is in contrast with the report of Erhabor *et al*., [24] who found a statistical significance between the marital status of blood donors and HCV reaction. Considering the occupational groups, a high prevalence of HCV antibodies was observed among the students (3.64%) followed by traders (8.70%). However, there was no statistical significant relationship ($P > 0.05$) between occupational status and the viral infection. The high prevalence might be due to exposure to environments where different people with various cultural background interact. The distribution of HCV antibody based on educational status showed that higher prevalence was recorded among those with tertiary education (14.71%), followed by those with non-formal

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education (11.11%) and secondary school level (2.56%). However, there was no statistical significant association (P > 0.05) between educational level and HCV positivity. The high prevalence among those with tertiary educational level and non-formal education might be due to low public awareness about the routes of transmission of the infection.

Among the pregnant women, the seroprevalence rate of HCV antibodies was 0.36% (1/279) (Table 2). This rate is low as compared with United States and other countries where they have reported the seroprevalence rate of HCV antibodies among the pregnant women to be between 0.7% and 12.3% [17, 39, 40]. In Nigeria, this value was nearly similar with a 0.5% prevalence recorded by Buseri *et al*., [29] among the study subject in Yenogoa, Bayelsa State and 0.5% seropositivity in pregnant women attending the University of Abuja teaching hospital, Gwagwalada [41]. In contrast, the finding in this study is much lower than anti-HCV antibody seroprevalence rates of 3.0% recorded by Osazuwa *et al*., [42] in Abaji Area Council, Abuja, 3.6% reported by Ugbebor *et al*., [43] among pregnant women attending the antenatal clinic of the University of Benin teaching hospital, 5.0% and 14.9% recorded in Faith me diplex hospital in Benin City and Enugu respectively [44]. The lower prevalence of anti-HCV antibody reaction in this study is ascribed to the difference in sample size, less exposure to blood transfusion and other possible risk factors.

As shown in Table 4, the highest prevalence of anti-HCV antibody was among the age range 36-45 years. This observation is consistent with the report of Duru *et al*., [44] who recorded high seroprevalence of HCV among the women in the age group 30-34 years, which are within the age range of this study. It is in contrast with the report of Ashok *et al*., [45] who recorded the

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highest prevalence in the age group 21-25 years. The difference may be due to less number of subjects in the older age group as compared with the age group of this study. However, there was no statistical significant association between age and HCV infection (P > 0.05).

Socio-demographic variables like age, marital and educational status, occupation and parity as well as risk factors such as previous history of blood donation, blood transfusion, previous surgery and alcohol intake were not significantly associated with the risk of HCV infection (*P* > 0.05) (Tables 3 and 4). This observation is consistent with the reports of Molla *et al*., [11], Oladeinde *et al*., [5] Yohannes *et al*., [21] and Esan *et al*., [46]. However, it is difficult to rule out the associations of these variables totally with HCV infection. This observation disagrees with the report of Rashed *et al*., [17] who identified positive correlation of the risk factors, such as history of blood transfusion, previous surgery, and high risk of occupation with increased risk of HCV infection.

Conclusions

A low prevalence rate of 1.79% was recorded among the blood donors and pregnant women attending Bowen University Teaching Hospital, Ogbomoso. No statistical significant association was observed in acquisition of HCV infection and socio-demographics among the two groups. Previous history of blood donation, blood transfusion, surgery procedure and alcohol intake were not found as important indicators of HCV infection among the study groups in this study. The following should be encouraged: Primary prevention activities, including screening and testing of blood, adequate sterilization of reusable materials such as surgical or dental instruments, risk-

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reduction counselling and services, implementation and maintenance of infection-control practices; Secondary prevention activities including identification, counselling, and testing of persons at risk, medical management of infected persons, surveillance and research to monitor disease trends and the effectiveness of prevention activities to develop improved prevention methods, as well as professional and public education. Molecular characterization should be carried out to determine the prevalence of HCV in future study.

Declaration of interest

All the authors declare that they have no commercial or any other association that might pose a conflict of interest for the manuscript enclosed.

Authors' contributions

AEH collected data and carried out the serological assays. OJK conceived the study. OEK reviewed the manuscript. EG performed the statistical analysis. All authors read and approved the final manuscript for publication.

References

Simmonds P: The origin and evolution of hepatitis viruses in humans. *J Gen Virol* 82, 693–712 (2001).

Ndjomou J, Pybus OG, Matz B. Phylogenetic analysis of hepatitis C virus isolates indicates a unique pattern of endemic infection in Cameroon. *J Gen Virol* 84, 2333-2341 (2003).

Accepted Manuscript

Moradpour D, Penin F. Hepatitis C virus proteins: from structure to function. *Curr Top Microbiol Immunol* 369, 113-142 (2013).

Bala JA, Kawo AH, Mukhtar MD, Sarki A, Magaji N, Aliyu A, Sani MN. Prevalence of hepatitis C virus infection among blood donors in some selected hospitals in Kano, Nigeria. *International Research J Microbiol* (IRJM) (ISSN: 2141-5463) 3, 217-222 (2012).

Oladeinde BH, Omoregie R, Olley M, Anunibe JA, Oladeinde OB. Hepatitis B and C viral infections among pregnant women in a rural community of Nigeria. *International Journal of Basic and Applied Virology*. 2012;1(1):01-05.

Lavanchy D. Global surveillance and control of hepatitis C. *Journal of Viral Hepatitis*. 1998;6:35-47.

World Health Organization (WHO). Hepatitis C Fact Sheet. April, 2014.

Lopez-Colombo A, Melendez-Mena D, Sedengo-Monge V, Camacho-Hernandez JR, Vazquez-Cruz E, Morales-Hernandez ER, Sosa-Jurado F, Marquez-Dominguez L, Santos-Lopez G. Hepatitis C virus infection in patients and family members attending two primary care clinics in Puebla, Mexico. Annals of Hepatology. 2013;13(6):746-752.

Halim NK, Ajayi OI. Risk factors and seroprevalence of hepatitis C antibody in blood donors in Nigeria. *East Afr Med J*. 2000;**77**: 410-2.

Afolabi AY, Abraham A, Oladipo EK, Fagbami AH. Hepatitis C virus in potential blood donors in Ibadan, Nigeria. *Global Advanced Research Journal of Microbiology*. 2012;1(9):155-159.

Accepted Manuscript

Molla S, Munshea A, Nibret E. Seroprevalence of hepatitis B surface antigen and anti-HCV antibody and its associated risk factors among pregnant women attending maternity ward of Felege Hiwot Referral Hospital, northwest Ethiopia: a cross-sectional study. *Virology Journal*. 2015;12:204.

Babatunde OM, Onoja MA, Israel E, Faneye AO, Ado UA, Onoja B. Seroprevalence rates of hepatitis C virus (HCV) antibody and hepatitis B virus surface antigen (Hbsag) in blood donors in a Southwestern Nigerian city. *Journal of Immunoassay and immunochemistry*. 2015;36(1):91-99.

Nwankwo E, Momodu I, Umar I, Musa B, Adeleke S. Seroprevalence of major blood-borne infections among blood donors in Kano, Nigeria. *Turk J Med. Sci.* 2012;42(2):337-341.

Diro E, Alemu S, Yohannes A. Blood safety and prevalence of transfusion transmissible viral infections among blood donors at Red Cross Bank in Gondar University Hospital. *Ethiop. Med. J*. 2008;46:7-13.

Mwambungu A, Siulapwa N, Mampi I. Seroprevalence of hepatitis B and C virus in blood donors at Lewanika General Hospital, Mongu Western Province, Zambia. *International Journal of Medical and Health Research*. 2015;1(1):80-84.

El Damaty SI, Hassan SK, Mohammed MK, El Hosini M, Rekacewicz C, Fontanet A. Surveillance system for HCV infection: testing a model based on blood banks. *J Egypt Public Health Assoc*. 2007;82:451-471.

Accepted Manuscript

Rashed M, Reham S, Abdel-Raouf O, Khattab A, Mohamed A, Sahar A. Hepatitis C virus infection and its vertical transmission in a sample of Egyptian pregnant women. *Journal of American Science*. 2015;11(1):40-46.

Ampofo W, Nii-Trebi N, Ansah J, Abe K, Naito H, Aidoo S. Prevalence of blood-borne infectious diseases in blood donors in Ghana. *J Clin Microbiol*. 2002;40:3523-3525.

Batina A, Kabemba S, Malengena R. Infectious markers among blood donors in Democratic Republic of Congo. *Rev. Med. Brux*. 2007;28:145-149.

Mantee MIN, Magesa PM, Lyamuya EF. Seroprevalence of human immunodeficiency virus, hepatitis B and C viruses and syphilis infection among blood donors at the Muhimbili National Hospital in Dar Es Salaam, Tanzania. *BMC Public Health*. 2006;6:21.

Yohannes Z. Wondemagegn M, Mulat Y, Bayeh A. Sero-prevalence and risk factors of hepatitis c virus infection among pregnant women in Bahir Dar city, Northwest Ethiopia: a cross sectional study. *Pan African Medical Journal*. 2015;21:158.

Vardas E, Sitas F, Seidel K, Castelinq A, Sim J. Prevalence of hepatitis C virus antibodies and genotypes in asymptomatic, first-time blood donors in Namibia. Bulletin of the World Health Organization. 1999;77(12):970-971.

Maha MMF, Wafa IE. Seroprevalence of hepatitis C virus among blood donors attending Omdurman teaching hospital. *European Academic Research*. 2015;3(2):1597-1608.

Erhabor O, Yakubu A, Abubakar AW, Buhari H, Okwesili A, Onuigwe FU, et al. The risk of transfusion-transmissible hepatitis C infection among blood donors in Sokoto, North Western Nigeria. *Asian Journal of Science and Technology*. 2015;6(2):1051-1057.

Accepted Manuscript

Olokoba AB, Badung LH, Abdulrahman MB, Salawu FK, Danburam A, Aderibigbe S, et al. Hepatitis C virus infection in Nigerians with diabetes mellitus. *African Journal of Scientific and Industrial Research*. 2010;1(2):135-138.

Ejele OA, Nwauche CA, Erhabor O. Seroprevalence of hepatitis C virus in the Niger Delta of Nigeria. *Niger Postgrad Med J*. 2006;13:183-6.

Adekeye AM, Chukwuedo AA, Zhakom PN, Yakubu RS. Prevalence of Hepatitis B and C among Blood Donors in Jos South LGA, Plateau State, Nigeria. *Asian Journal of Medical Sciences*. 2013;5(5):101-104.

Mutimer DJ, Olomu A, Skidmore S. Viral hepatitis in Nigeria- sickle cell disease and commercial blood donors. *Quaterly Journal of Medicine*. 1994;87:407-411.

Buseri FI, Muhibi MA, Jeremiah Z.A. Sero-epidemiology of transfusion-transmissible infectious diseases among blood donors in Osogbo, South-West Nigeria. *Blood Transfusion*. 2009;7(4):292-299.

Egah DZ, Mandong BM, Iya D, Gomwalk NE, Audu ES, Banwat EB, Onile BA. Hepatitis C virus antibodies among blood donors in Jos, Nigeria. *Annals of African Med.* 2004;391:35-37.

Adejare EOA, Oyegbami TO, Azenabo A, Abioye AO, Ahmed IB, Awokola OT, Odeyemi AA, Adewale CO. Seropositivity of hepatitis B and C among blood donors in private laboratories in Lagos, Nigeria. *Journal of Biology, Agriculture and Healthcare*. 2015;5(3).

Achinge GI, Malu AO, Mbaave PT, Bitto TT, Shaahu VN, Mohammed H, Misauno MA. Prevalence of hepatitis C in Makurdi, North Central Nigeria. *IOSR Journal of Dental and Medical Sciences*. 2013;7(5):06-10.

Accepted Manuscript

Obienu O, Nwokediuko S, Malu A, Lesi F. Risk factors for hepatitis C virus transmission obscure in Nigerian patients. *Gastroenterology Research and Practice*. 939673, 2011;4 pages.

Mboto CI, Andy I.E, Eni OI, Jewell AP. Prevalence, socio-demographic characteristics and risk factors for hepatitis C infection among pregnant women in Calabar municipality, Nigeria. *Hepat Mon*. 2010;10(2):116-120.

Muktar HM, Suleiman AM, Jones M. Safety of blood transfusion: prevalence of hepatitis B surface antigen in blood donors in Zaria, Northern Nigeria. Nigerian Journal of Surgical Research. 2005;7(3&4):290-292.

Malhotra P, Malhotra V, Malhotra N, Chugh A, Chaturvedi A. Epidemiological profile of hepatitis C patients at India's New Hub- Haryana. *Advanced Research in Gastroenterology and Hepatology*. 2015;1(1):ARGH.MS.ID.555554.

Inyama PU, Uneke CJ, Anyanwu GI, Njoku OM, Idoko JA. Prevalence of HBV among Nigerian patients with HIV infection. *Journal of Health and Allied Sciences.* 2005;2:2.

Sule WF, Odama LE, Daniel AL, Faneye AO, Okonko IO. Prevalence of anti-hepatitis C virus in patients attending Federal medical Centre (FMC) Lokoja, Kogi State, Nigeria. *World Applied Sciences Journal*. 2009;7(3):371-377.

Landes M, Newell ML, Barlow P. Hepatitis B or hepatitis co-infection in HIV-infected pregnant women in Europe. *HIV Medicine*. 2008;9:526-534.

Menendez C, Sanchez-Tapias JM, Kahigwa E. Prevalence of mother-to-child transmission of hepatitis viruses B, C, and E in Southern Tanzania. 1999;58(3):215-220.

Accepted Manuscript

Agarry OO, Zamani LG. Prevalence of hepatitis B znd C virus in ante-natal patients in Gwagwalada-Abuja, Nigeria. *Report and Opinion*. 2010;2:7.

Osazuwa F, Obinna OV, Anikwe, FC. Sero-epidemiology of human immunodeficiency virus, hepatitis B and C among pregnant women in rural communities of Abaji Area Council, Nigeria. *TAF Preventive Medicine Bulletin*. 2012;11(4):431-438.

Ugbebor O, Aigbirior M, Osazuwa F, Enabudoso E, Zabayo O. The prevalence of hepatitis B and C viral infections among pregnant women. *North Am J Med Sci*. 2011;3:238-241.

Duru MU, Aluyi HAS. Anukam KC. Rapid screening for co-infection of HIV and HCV in pregnant women in Benin City, Edo State, Nigeria. *Afr Health Sci*. 2009;9:137-142.

Ashok K, Sharma KA, Gupta RK, Kar P, Chakravarti, A. Prevalence and risk factors for hepatitis C virus among pregnant women. *Indian J Med Res*. 2007;126:211-215.

Esan AJ, Omisakin CT, Ojo-Bola T, Owoseni MF, Fasakin KA, Ogunleye AA. Sero-Prevalence of Hepatitis B and Hepatitis C Virus Co-Infection among Pregnant Women in Nigeria. Am J Biomed Res. 2014;2(1):11–5.

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Table 1: Socio-demographic characteristics of the study groups.

		BLOOD DONORS		PREGNANT WOMEN	
Parameter		Frequency	Percent (%)	Frequency	Percent (%)
		N = 141		N = 138	
Sex	Male	123	87.23	-	-
	Female	18	12.77	138	100.00
Age group (years)	16-25	61	42.36	21	15.22
	26-35	44	31.21	94	68.12
	36-45	23	16.31	23	16.67
	46-55	12	8.51	1	0.72
	56-65	1	0.71	0	0.00

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Marital status	Single	78	55.32	0	0.00
	Monogamy	56	39.72	132	95.65
	Polygamy	7	4.96	6	4.35
Educational status	Tertiary	68	48.23	76	55.07
	Secondary	39	27.66	38	27.54
	Primary	16	11.35	10	7.25
	Others	0	0.00	12	8.70
	None	18	12.77	2	1.45
Occupation	Student	55	39.01	13	9.42
	Artisan	36	25.53	21	15.22

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Trading	23	16.31	48	34.78
Civil servant	23	16.31	46	33.33
Others	4	2.84	1	0.72
None	0	0.00	9	6.52

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Table 2: Seroprevalence of HCV infections among the study groups.

STUDY GROUPS	STATUS	TOTAL (%)
BLOOD DONORS	Reactive	4 (2.84)
	Non-reactive	137 (97.16)
	Total	141 (100.00)
PREGNANT WOMEN	Reactive	1 (0.72)
	Non-reactive	137 (99.28)
	Total	138 (100.00)

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Table 3: Seroprevalence of HCV in relation to socio demographic characteristics among the blood donors.

Parameter	Level	HCV status				Df	Chi-square	p-value
		Reactive		Non-reactive				
		Count	Percentage (%)	Count	Percentage (%)			
Age (years)	16-25	3	4.92	58	95.08	4	2.813	0.590
	26-35	0	0.00	44	100.00			
	36-45	1	4.35	22	95.65			
	46-55	0	0.00	12	100.00			
	56-65	0	0.00	1	100.00			
Sex	Male	4	3.25	119	96.75	1	0.641	0.557
	Female	0	0.00	18	100.00			

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Marital status	Single	3	3.85	75	96.15	2	0.717	0.699
	Monogamy	1	1.79	55	98.21			
	Polygamy	0	0.00	8	100.00			
Educational status	Tertiary	1	14.71	67	98.53	3	5.009	0.171
	Secondary	1	2.56	38	97.44			
	Primary	0	0.00	16	100.00			
	None	2	10.53	17	89.47			
Occupation	Student	2	3.64	53	96.36	4	4.831	0.305
	Artisan	0	0.00	36	100.00			
	Trading	2	8.70	21	91.30			
	Civil servant	0	0.00	23	100.00			

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Others	0	0.00	4	100.00

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Table 4: Seroprevalence of HCV in relation to socio demographic characteristics among the pregnant women.

Parameter	Level	HCV status				Df	Chi-square	p-value
		Reactive		Non-reactive				
		Count	Percentage (%)	Count	Percentage (%)			
Age (years)	16-25	0	0.00	21	100.00	3	5.311	0.150
	26-35	0	0.00	94	100.00			
	36-45	1	4.55	21	95.45			
	46-55	0	0.00	1	100.00			
Marital status	Monogamy	1	0.75	131	99.24	1	0.046	0.831
	Polygamy	0	0.00	6	100.00			
Educational	Tertiary	1	1.32	75	98.53	4	0.822	0.936

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status	Secondary	0	0.00	38	100.00			
	Primary	0	0.00	10	100.00			
	None	0	0.00	2	100.00			
	Others	0	0.00	12	100.00			
Occupation	Student	0	0.00	13	100.00	5	2.015	0.847
	Artisan	0	0.00	21	100.00			
	Trading	0	0.00	48	100.00			
	Civil servant	1	2.17	45	97.83			
	Others	0	0.00	1	100.00			
	None	0	0.00	9	100.00			
Parity	Multigravida	1	2.00	49	98.00	2	1.773	0.412

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Primigravida	0	0.00	79	100.00
First issue	0	0.00	9	100.00

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Table 5: Association of potential risk factors and hepatitis C virus infection among the apparently healthy individuals

Variables	HCV Status		COR (Cl)	AOR (Cl)	*P*-value
	Positive N (%)	Negative N (%)			
History of blood donation					
Yes	1(2.0)	48(98.0)	0.148(0.062-6.098)	1.571(0.157-15.706)	0.682
No	3(3.3)	89(96.7)	1	1	
Surgical procedure					
Yes	0(0.00)	10(100.00)	0.000	27138083.69	0.999
No	4(3.05)	127(96.95)			
Blood transfusion					
	0(0.00)	7(100.00)	0.000	18837199.36	0.999

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Yes	4(2.99/)	130(97.01)			
No					
Alcohol intake					
Yes	0(0.00)	30(100.00)	0.000	51152069.28	0.998
No	4(3.60)	107(96.40)			

Key: COR = Crude odd ratio, AOR = adjusted odd ratio, P-value < 0.05, 1 = reference value.

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Table 6: Association of potential risk factors and hepatitis C virus infection among pregnant women

Variables	HCV Status		COR (Cl)	AOR (Cl)	*P*-value
	Positive N (%)	Negative N (%)			
History of blood donation					
Yes	0(0.00)	6(100.00)	0.000	15097895.802	0.999
No	1(0.76)	131(99.24)	1		
Surgical procedure					
Yes	0(0.00)	22(100.00)	0.000	12521128.428	0.998
No	1(0.86)	115(99.14)			
Blood transfusion					
	0(0.00)	4(100.00)	0.000	2780628.411	0.999

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Yes	1(0.75)	133(99.25)			
No					
Alcohol intake					
Yes	0(0.00)	1(100)	1.000	15097898.387	1.000
No	1(0.73)	136(99.27)			

Key: COR = Crude odd ratio, AOR = adjusted odd ratio, P-value < 0.05, 1 = reference value.

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